Mobile voting system for zetech university

SYSTEM DOCUMENTATION

SUBMITTED BY,

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A SYSTEM DOCUMENTATION SUBMITTED IN PARTIAL FULFILMENT FOR THE AWARD OF DIPLOMA IN SOFTWARE ENGINEERING BY ZETECH UNIVERSITY

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DECLARATION

I, George Njoroge, hereby declare that this project titled Mobile Voting System for Zetech University and the accompanying system documentation are my original work. I confirm that this project has not been submitted for any other award or qualification at Zetech University or any other institution. The work presented herein has been independently researched, designed, and developed by me, and all sources of information have been duly acknowledged.

Student Name: George Njoroge

Student Signature: ______________________

Date: ______________________

Supervisor's Confirmation

This project has been submitted for examination with my approval as the appointed supervisor.

Supervisor Name: Nicholas Kavoi

Supervisor Signature: ______________________

Date: ______________________

DEDICATION

I dedicate this project to Almighty God for His guidance and strength throughout this journey; to my beloved parents, whose unwavering support, encouragement, and sacrifices have made my education possible, and whose belief in me has been my greatest motivation; to my lecturers and mentors at Zetech University, who have imparted knowledge and shaped my skills in software engineering; and to all students who aspire to leverage technology for better governance and participation in academic institutions.

ABSTRACT

This project presents the development of a web-based hierarchical student voting application for Zetech University, designed to overcome the inefficiencies of traditional manual voting such as time consumption, errors, and inaccessibility for off-campus students which previously led to low participation and delayed results. Developed over seven weeks following the software development lifecycle, the system uses HTML5, CSS3, and JavaScript for a responsive frontend, with Firebase Authentication for secure student verification, Firebase Realtime Database for backend storage, and Firebase Cloud Functions for automated vote validation, one-vote-per-user enforcement, and real-time tallying. It implements a two-tier hierarchical structure (departmental and executive elections) with role-based access control for students, department representatives, and administrators, offering secure authentication, real-time results, automated vote counting, and election management. Deployed via Firebase Hosting with GitHub integration, testing confirmed accurate vote enforcement, data integrity, and timely results, successfully achieving an accessible, secure, and efficient platform that enhances student participation and transparency in university elections.

DEFINITION OF KEY TERMS

Administrator: A system user with elevated privileges responsible for creating and managing elections, registering candidates, and overseeing the voting process.

Authentication: The process of verifying a user's identity before granting access to the system, implemented in this project through Firebase Authentication using university credentials.

Candidate: An individual registered to contest for a specific position (either department representative or executive council member) in an election.

Database: A structured collection of data stored electronically. In this project, Firebase Realtime Database stores users, elections, candidates, and votes.

Department Representative: An elected student who represents their academic department and has voting privileges in executive-level elections.

Election: A formal voting process where students elect representatives for departmental or executive positions.

Executive Election: A university-wide voting process conducted by department representatives to elect members of the student council, including positions such as President and Secretary General.

Firebase: A Backend-as-a-Service (BaaS) platform by Google that provides authentication, real-time database, and hosting services used in this project.

Frontend: The client-side interface of the web application that users interact with, developed using HTML, CSS, and JavaScript.

One-Vote Enforcement: A system mechanism that ensures each user can vote only once per election position.

Real-Time Results: Election outcomes that are instantly updated and displayed as votes are cast and counted.

Role-Based Access Control: A security mechanism that restricts system access based on user roles (student, department representative, administrator).

Vote: The recorded choice made by a user for a specific candidate or position.

Voter Eligibility: The validation process that confirms a user's right to participate in a specific election based on their department affiliation or role.

Web Application: An interactive software application accessed through a web browser, which forms the foundation of this voting system.

ABBREVIATIONS AND ACRONYMS

Abbreviation	Full Meaning

Abbreviation	Full Meaning
API	Application Programming Interface
CLI	Command Line Interface
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
GitHub	A web-based platform for version control and collaboration
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IDE	Integrated Development Environment
JS	JavaScript
MFA	Multi-Factor Authentication
PC	Personal Computer
PWA	Progressive Web Application
SDK	Software Development Kit
UI	User Interface
URL	Uniform Resource Locator

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CHAPTER ONE: PROJECT PLANNING AND ANALYSIS (WORKPLAN)

1.1 Statement of the Problem

Zetech University currently relies on traditional or semi-manual voting methods for student elections, which are often time-consuming, costly, and prone to errors and inefficiencies (Zetech University, n.d., Varsity Swears In New Student Council). These methods can result in delayed results, limited student participation, lack of transparency, and difficulties in verifying voter eligibility. Furthermore, physical voting processes are inconvenient for students who may be off-campus or unavailable during election periods. With the increasing adoption of mobile devices and web technologies among students, there is a growing need for a secure, efficient, and accessible digital voting solution. Studies indicate that digital voting systems improve participation, accuracy, and transparency in electoral processes (Kahle et al., 2020; MDPI, 2025). Therefore, this project seeks to develop a web-based voting application that enables students to vote electronically, ensures accurate vote tallying, and provides a transparent election process at Zetech University.

1.2 Study Justification

The study is justified by the need to modernize and streamline the student election process at Zetech University, as traditional voting methods face logistical challenges, delays in result processing, and limited participation from students unable to vote physically. A web-based voting application can provide the following benefits: increase accessibility by allowing students to vote from any location using mobile devices or web browsers; enhance efficiency through automated vote tallying that reduces administrative workload and speeds up result announcements; improve accuracy via digital validation that minimizes errors common in manual counting; ensure transparency with electronic records that provide verifiable and auditable election results; and promote inclusivity so that off-campus students and those with mobility constraints can participate fully. Implementing a web-based voting system supports institutional governance and student engagement while leveraging modern technologies to provide a reliable, secure, and accessible election process (MDPI, 2025; IJRASET, 2025).

1.3 System Objectives

1.3.1 General Objective

1. To develop a web-based hierarchical student voting application for Zetech University.

1.3.2 Specific Objectives

1. To design a responsive web-based voting interface.
2. To integrate the system with the university student database for authentication.
3. To implement role-based access control for students, representatives, and administrators.
4. To support departmental elections for electing representatives.
5. To support university executive elections conducted by department representatives.
6. To enforce one vote per user per election.
7. To automate vote counting and result generation.
8. To provide real-time election results through the web application.

1.4 Functional Requirements

1.5 Breakdown of Tools & Resources

User Role	Function	Description of User Activity in the Web Application
Student (Department Voter)	Login	Logs into the web application using university credentials. The system automatically verifies the student against the university database and confirms department affiliation.
	View Department Elections	Views active elections within their registered department.
	View Department Candidates	Views profiles of candidates contesting for department representative.

User Role	Function	Description of User Activity in the Web Application
	Cast Vote (Department Level)	Selects and submits a vote for one department representative.
	Vote Confirmation	Receives confirmation that the vote has been successfully recorded.
	View Results	Views published results after the departmental election is closed.
Department Representative (Delegate)	Login	Logs into the system with verified credentials. The system recognizes the user as an elected representative based on assigned role.
	View Executive Elections	Accesses university-level elections (e.g., President, Secretary General).
	View Executive Candidates	Views profiles of candidates contesting for main council positions.
	Cast Vote (Executive Level)	Submits a vote for executive council positions on behalf of the department.
	View Executive Results	Views published results after executive elections are closed.
Administrator	Create Department Elections	Configures and schedules elections for department representatives.
	Create Executive Elections	Configures university-level elections for main council positions.
	Manage Candidates	Registers, edits, or removes candidates for both election levels.
	Assign Representative Role	Assigns elected department representatives system privileges to participate in executive elections.

	Open/Close Voting Sessions	Controls when voting sessions begin and end.
	Monitor and Publish Results	Automatically tallies votes and publishes verified results.

1.5.1 Software Tools

- HTML, CSS, Java: Frontend and backend development of the web app.

- Android Studio: Testing mobile access and hybrid web views.

- Firebase Authentication: Manage secure login and voter verification.

- Firebase Realtime Database: Store user accounts, votes, and election data.

- Firebase Cloud Functions: Backend logic for vote validation and counting.

- GitHub: Version control and collaboration.

1.5.2 Hardware Resources

- Personal Computer: Coding, testing, documentation.

- Android Smartphone: Testing application accessibility and responsiveness.

1.5.3 Other Resources

- Internet Connection: Required for Firebase services and testing.

- Documentation Tool: Microsoft Word for project report.

- Operating System: Windows 10.

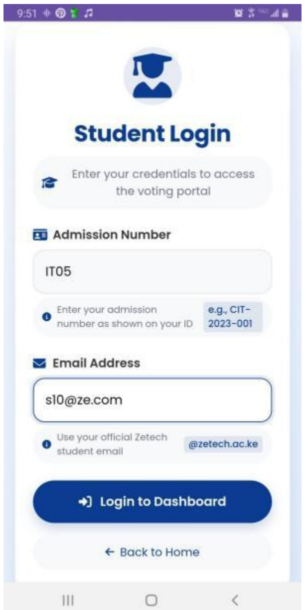
1.6 Project Schedule Breakdown

	PROJECT MILESTONES					
WEEKS	Project Planning & Analysis (System Documentation: Coverpage & Chapter One)	Project Design & Modeling (System Documentation Chapter Two)	Project Development & Testing (System Documentation Chapter Three)	Project Deploy m ent (System Documentation Chapter Three)	Final Touches of System Documentation (Preliminary Pages, Chapter Four & References)	Project Presen t ation
7-9 Jan						
12-16 Jan						
19-23 Jan						
26-30 Jan						
2-6 Feb						
9-13 Feb						
16-20 Feb						
23-27 Feb						
2-6 Mar						
9-13 Mar						
16-20Mar						
23-27 Mar						
3 April						

Table 1.6 Project Schedule Breakdown

Fig 2.2.1 Login Page

CHAPTER TWO: DESIGN AND MODELING



Introduction to Modelling

This chapter presents the designs and models developed to visualize the web-based student voting system before implementation. Modelling helped in understanding system structure, user interaction, data flow, and overall system behavior, reducing errors and improving efficiency during development.

2.1 User Interface Models

User interface models were designed to visualize how system pages would appear to users.

2.1.1 Login Page Design

The login page allows students and administrators to securely access the system.

Fig 2.2.1 Login Page

2.1.2 Voter Dashboard Design

Displays available elections and voting status.

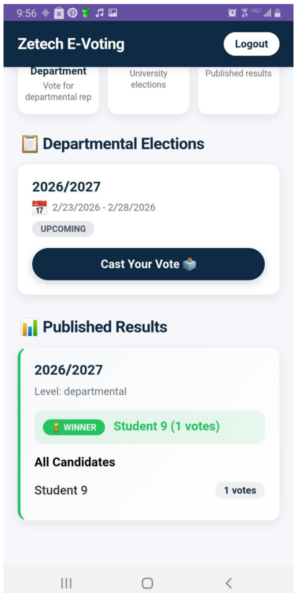


Fig 2.2.3 Voter Dashboard Interface

2.1.3 Voting Page Design

Allows voters to select and submit a candidate.

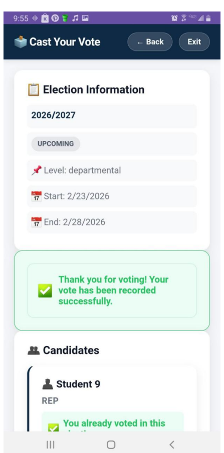


Fig 2.2.4 Voting Page

2.2.5 **Vote Confirmation Dialog**

Design Confirms vote before

final submission.

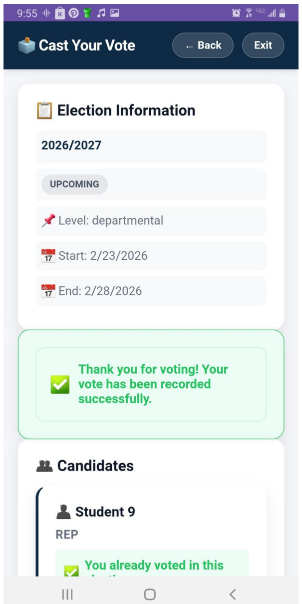


Fig 2.2.5 Vote Confirmation Dialog Design

2.2.6 **Results Page**

Design Displays

election results.

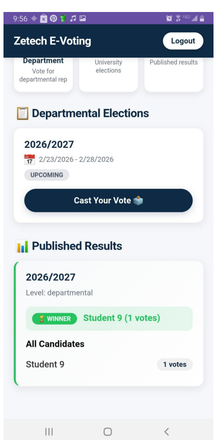


Fig 2.2.6 Election Results Page

2.2.7 Admin Dashboard Design

Used by administrators to manage elections.

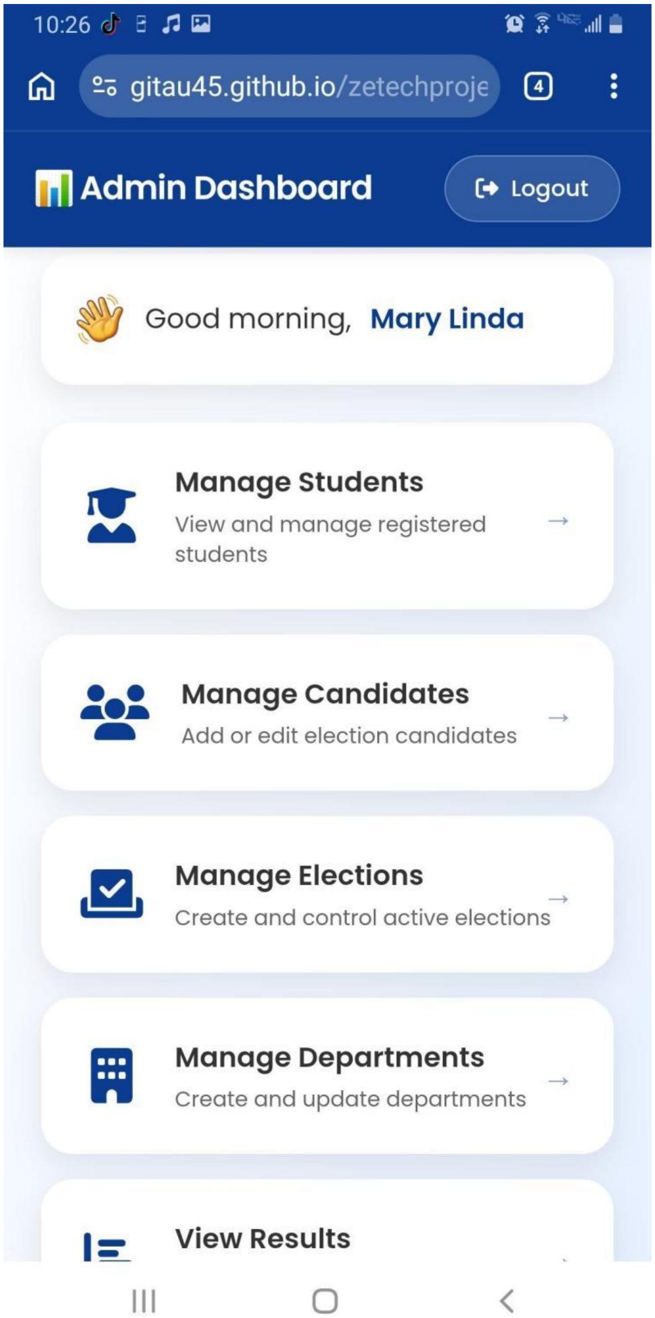


Fig 2.2.7 Admin Dashboard

2.2 **Logic Models**

2.2.1 Use Case Diagram

Shows interaction between users and the system.

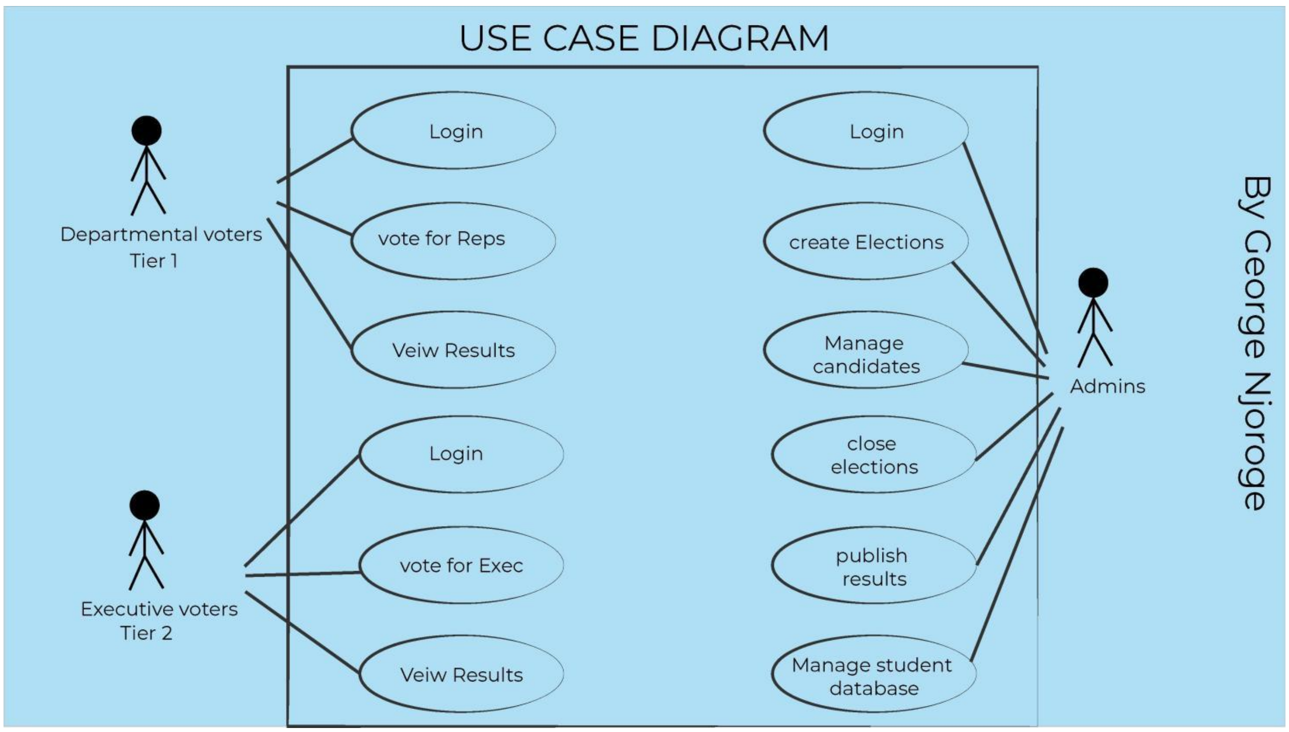


Fig 2.3.1 Use Case

2.2.2 **Login Process**

Flowchart Illustrates

login process logic.

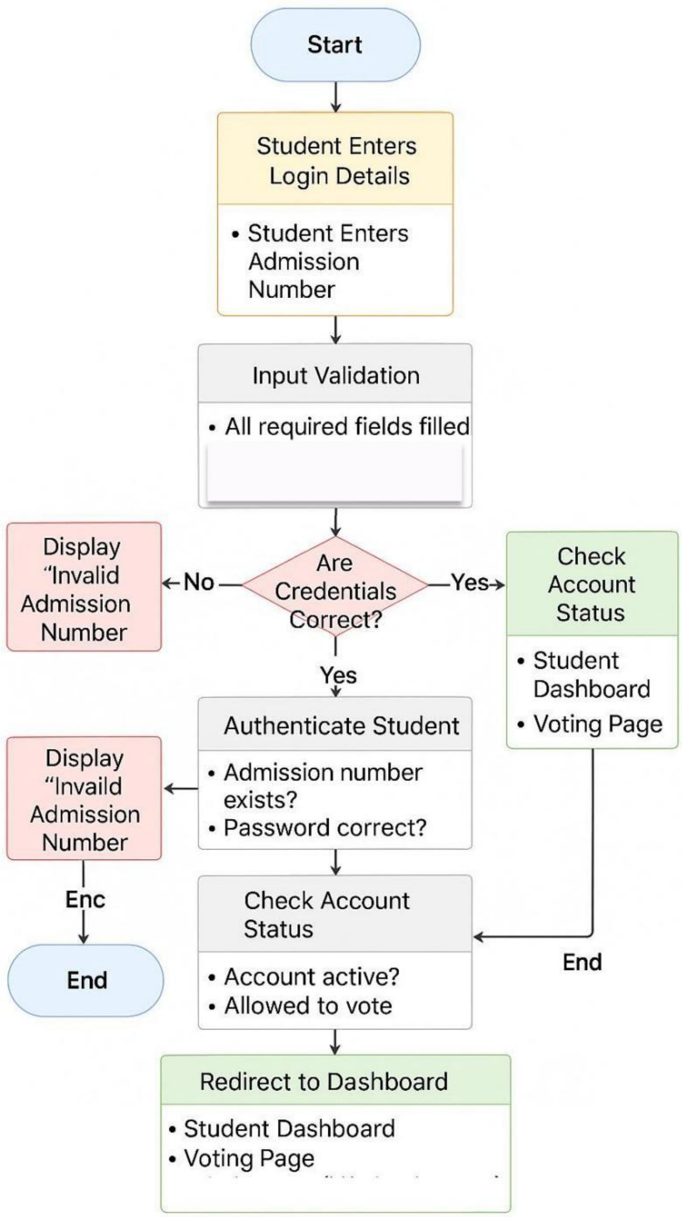


Fig 2.3.2 Login Process

2.2.3 **Elections creation Process**

Flowchart Shows steps followed

during voting.

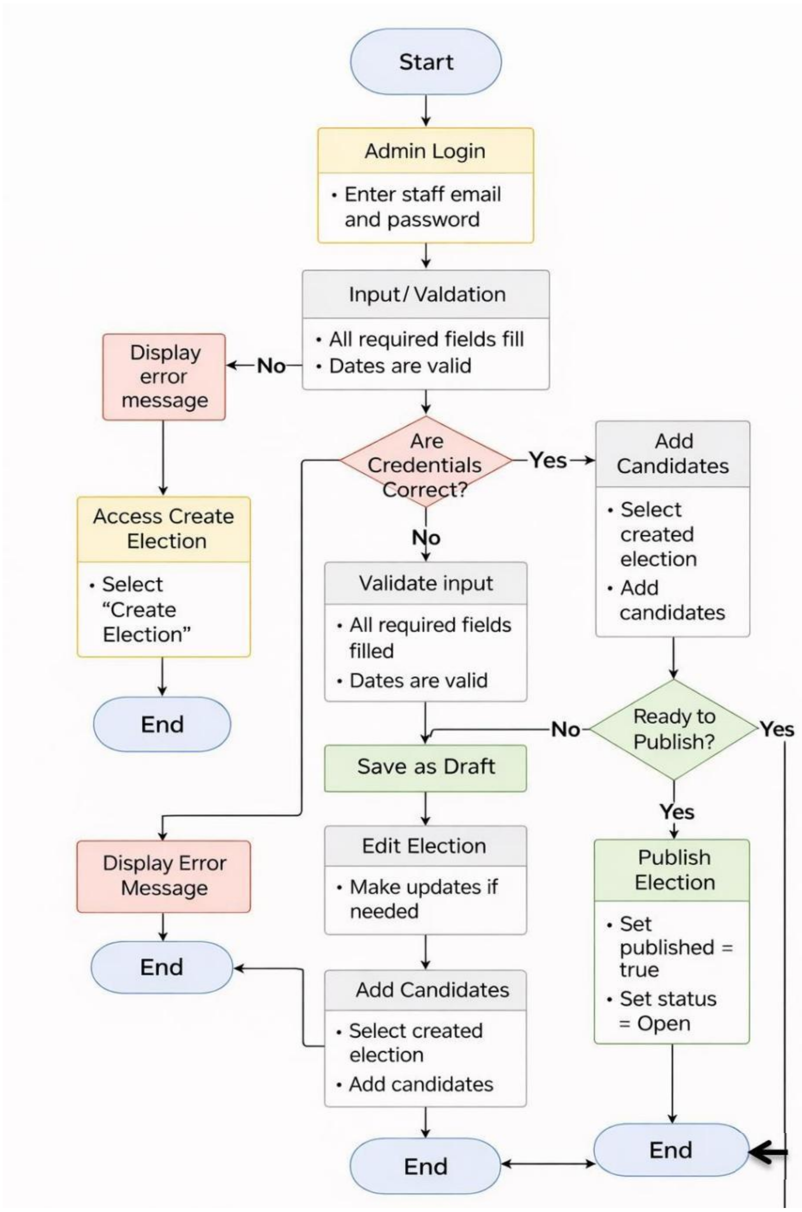


Fig 2.3.3 Elections creation Process

2.2.4 **Entity Relationship**

Diagram Shows database

structure.

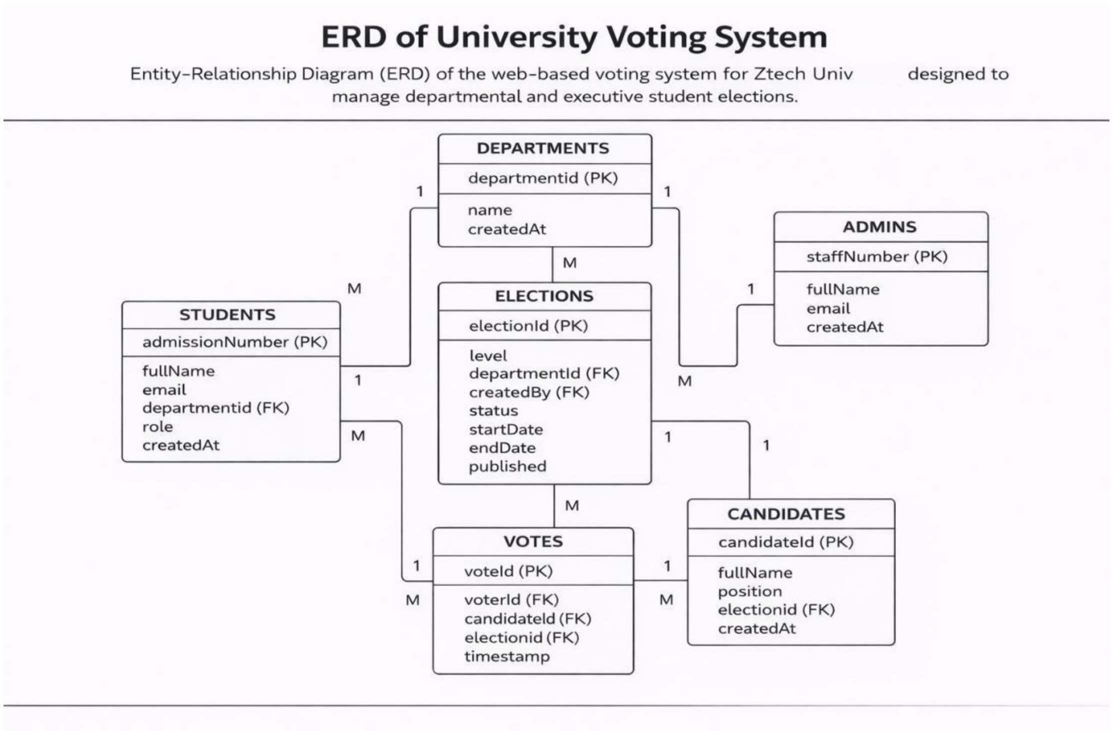


Fig 2.3.4 Entity Relationship Diagram

Relationship Summary

Department → Students = One-to-

Many Department → Elections =

One-to-Many Admin → Elections =

One-to-Many Election →

Candidates = One-to-Many

Election → Votes = One-to-Many

Student → Votes = One-to-Many

Candidate → Votes = One-to-Many

2.2.5 **Data Flow** Diagram

Shows dataflow

structure.

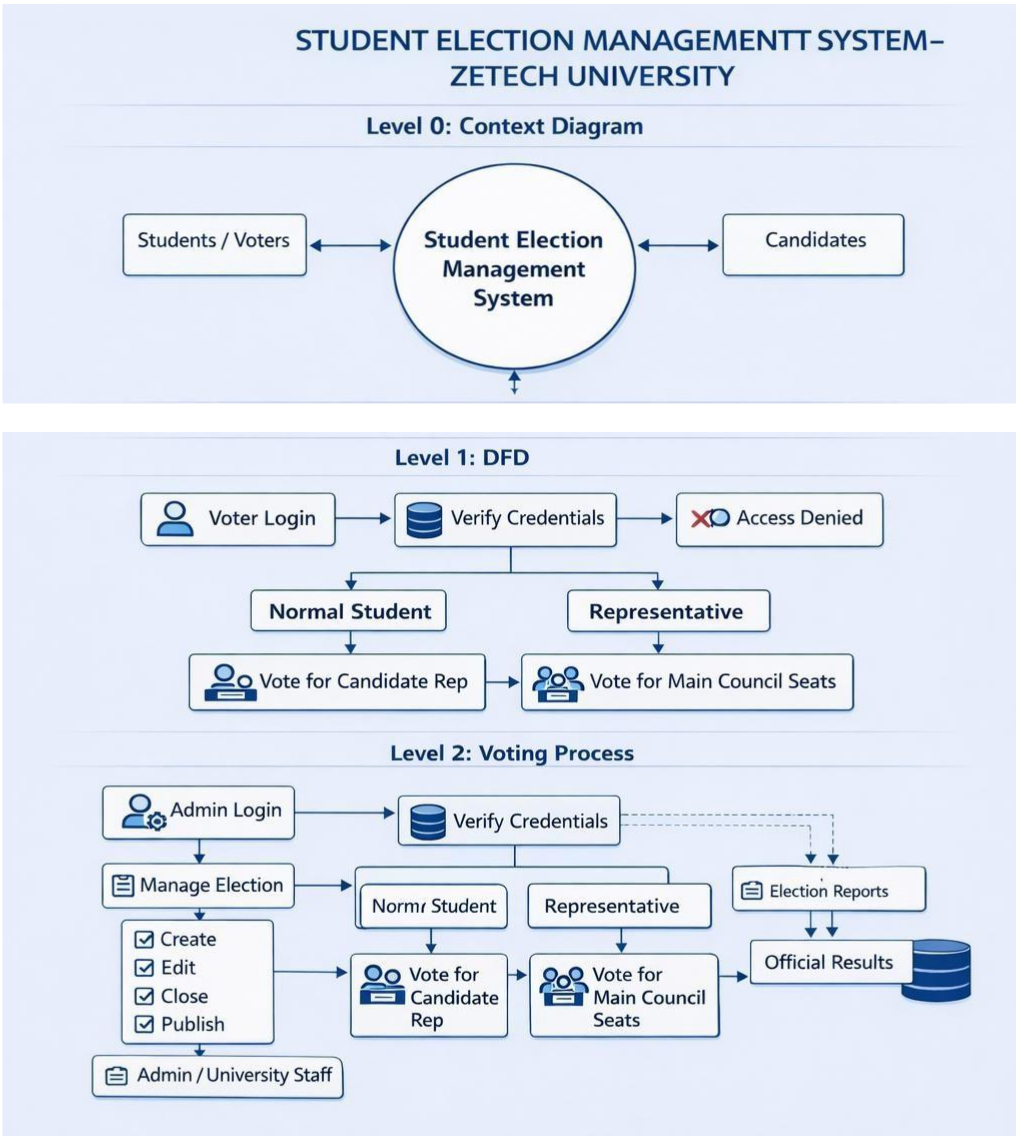


Fig 2.3.5 Data Flow Diagram

CHAPTER THREE: SYSTEM IMPLEMENTATION (DEVELOPMENT, TESTING AND DEPLOYMENT)

3.1 Introduction

This chapter presents the system implementation process of the Web-Based Voting Application for Zetech University. It outlines the development journey from environment setup, user interface design, backend logic implementation, database integration, testing procedures, and final system deployment. The chapter demonstrates how the theoretical designs discussed in earlier chapters were transformed into a fully functional web-based voting system.

3.2 User Interface Development

The user interface (UI) represents the front-end part of the system where students and administrators interact with the application. The interfaces were developed using HTML, CSS, and Java, and tested using modern web browsers and Android devices for responsiveness.

3.2.1 Login Page Development

This section explains the purpose of this interface, how it was developed, and its key features. The page was designed to be user-friendly, responsive, and secure.

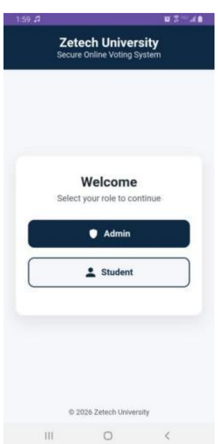


Figure: Screenshot of the interface.

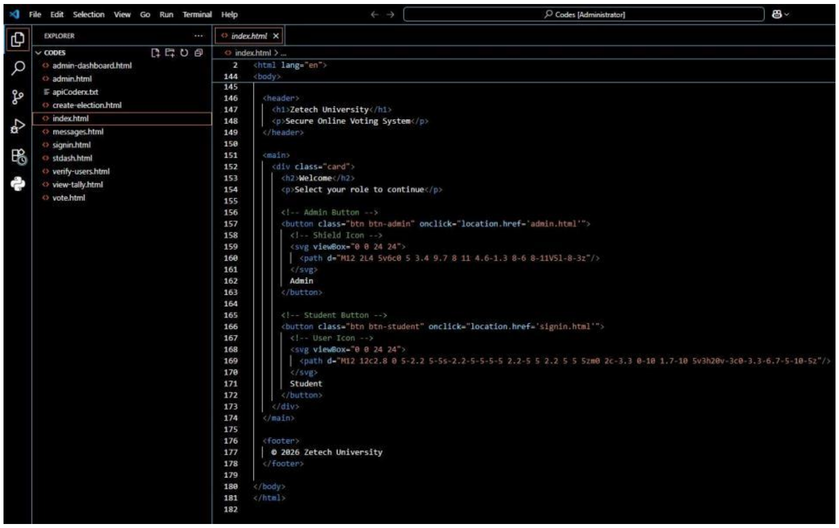


Figure: Screenshot of the source code used to develop the interface.

3.2.2 Student Registration Page Development

This section explains the purpose of this interface, how it was developed, and its key features. The page was designed to be user-friendly, responsive, and secure.

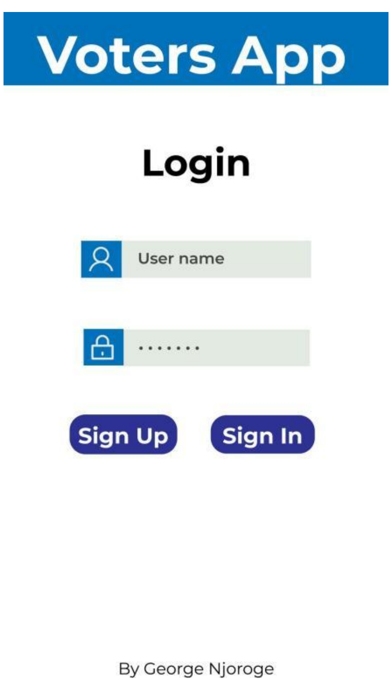


Figure: Screenshot of the interface.

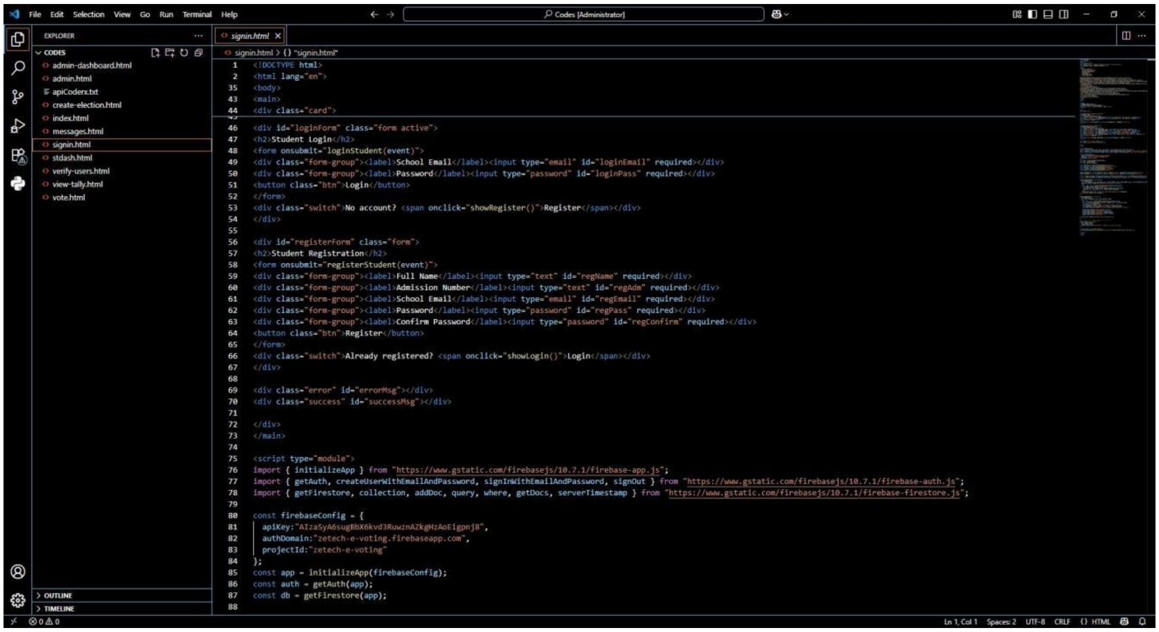


Figure: Screenshot of the source code used to develop the interface.

3.2.3 Student Voting Dashboard

This section explains the purpose of this interface, how it was developed, and its key features. The page was designed to be user-friendly, responsive, and secure.

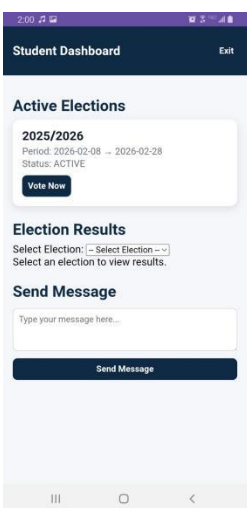


Figure: Screenshot of the interface.

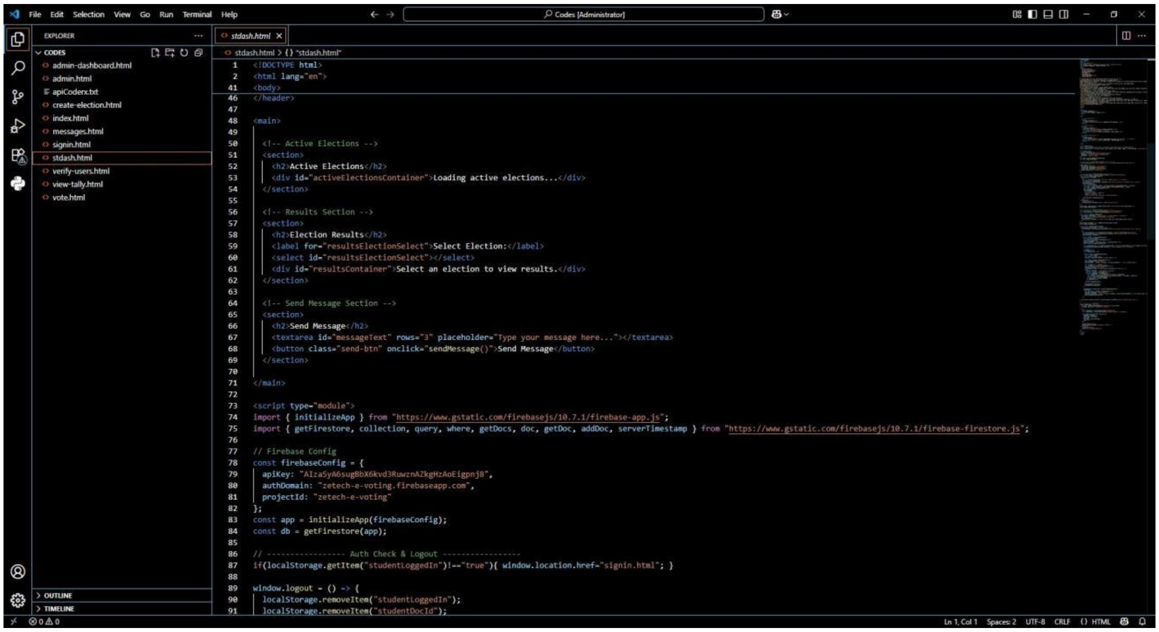


Figure: Screenshot of the source code used to develop the interface.

3.2.4 Admin Dashboard

This section explains the purpose of this interface, how it was developed, and its key features. The page was designed to be user-friendly, responsive, and secure.

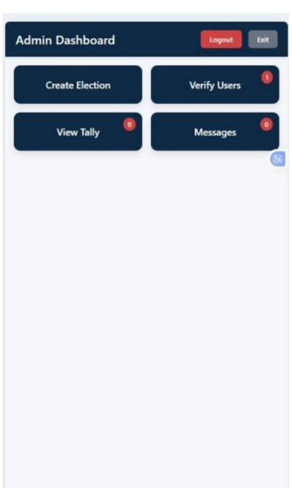


Figure: Screenshot of the interface.

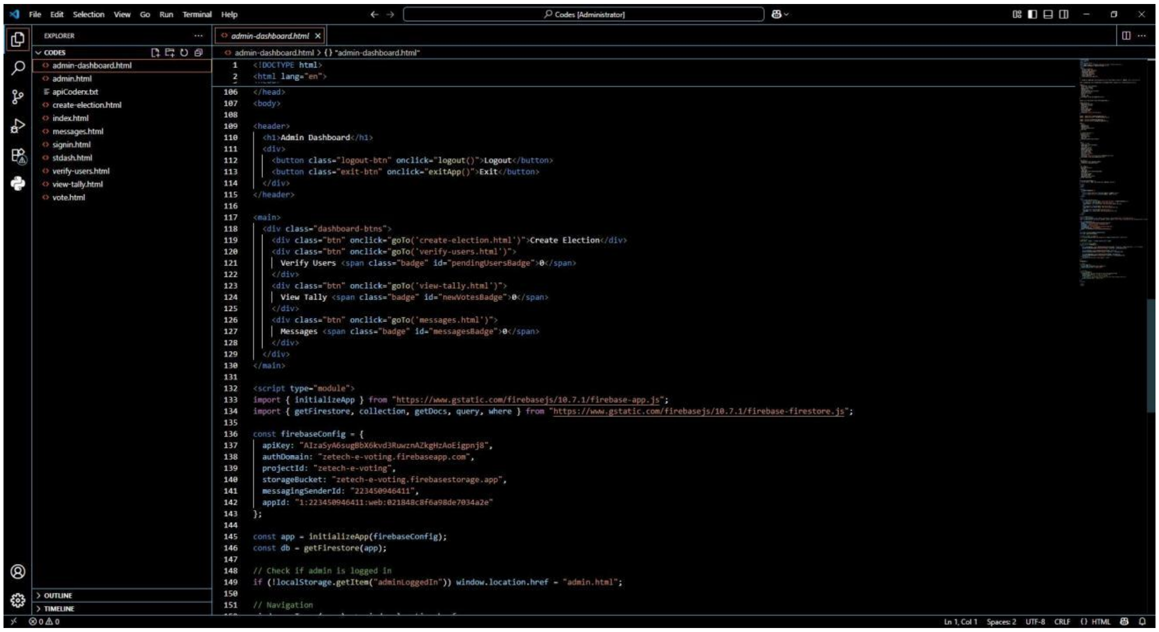


Figure: Screenshot of the source code used to develop the interface.

3.2.5 Election Results Page

This section explains the purpose of this interface, how it was developed, and its key features. The page was designed to be user-friendly, responsive, and secure.

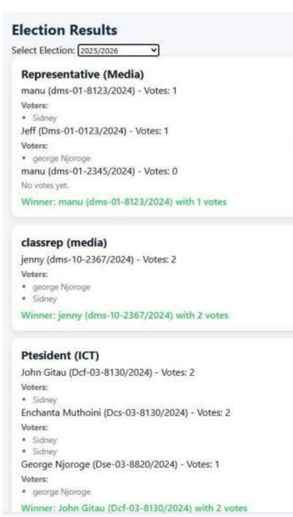


Figure: Screenshot of the interface.

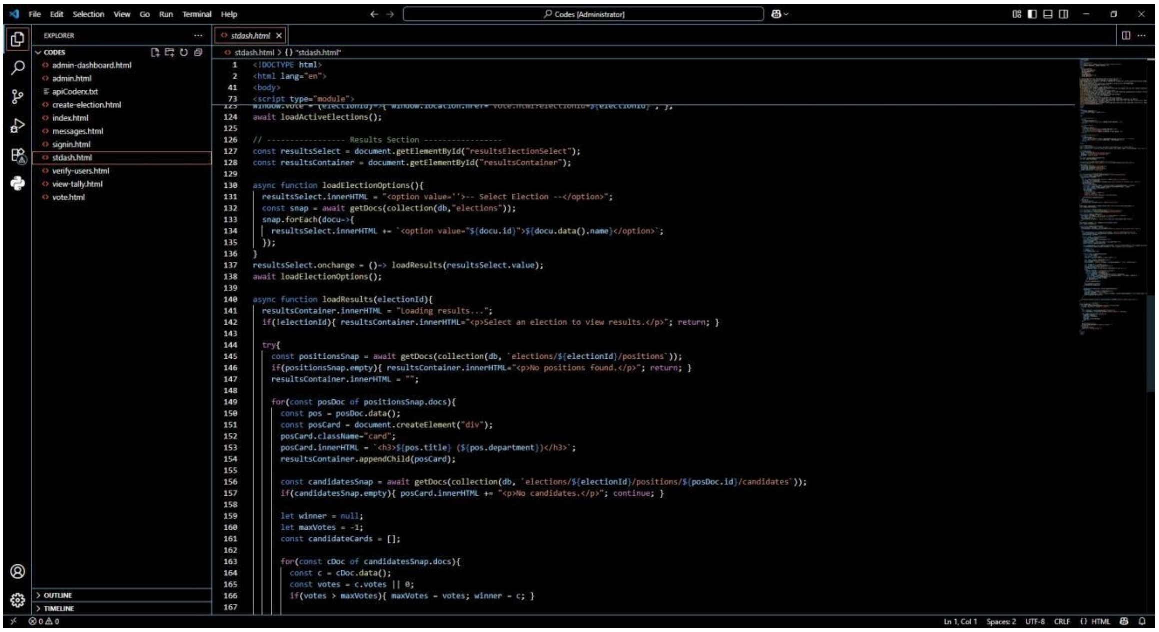


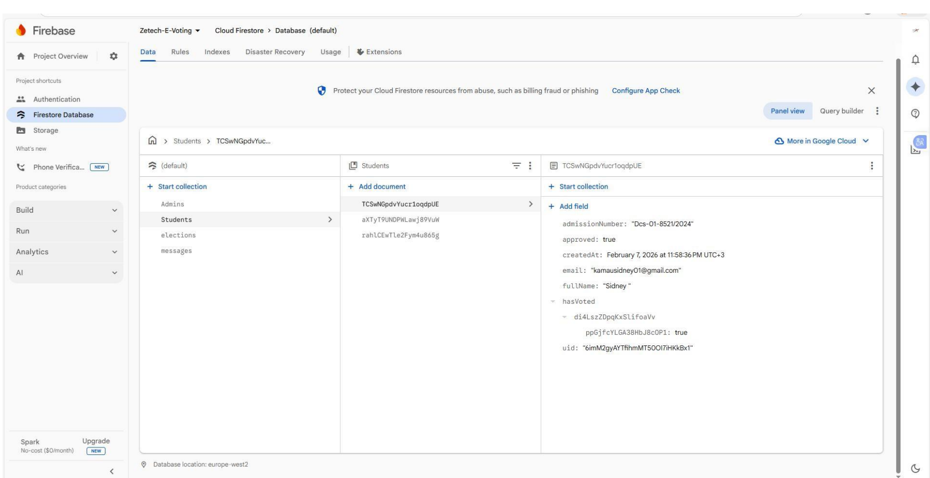
Figure: Screenshot of the source code used to develop the interface.

3.3 Logic Development

The logic side of the system handles data processing, authentication, vote validation, database interactions, and automated vote counting. Firebase Authentication, Firebase Realtime Database, and Cloud Functions were used to implement backend logic.

3.3.1 Login Validation Logic

This section explains how the system processes this functionality. The backend logic ensures security, accuracy, and reliability of the voting process.



re.

Figure: Screenshot of the logic implementation code.

3.3.2 Voter Eligibility Verification

This section explains how the system processes this functionality. The backend logic ensures security, accuracy, and reliability of the voting process.

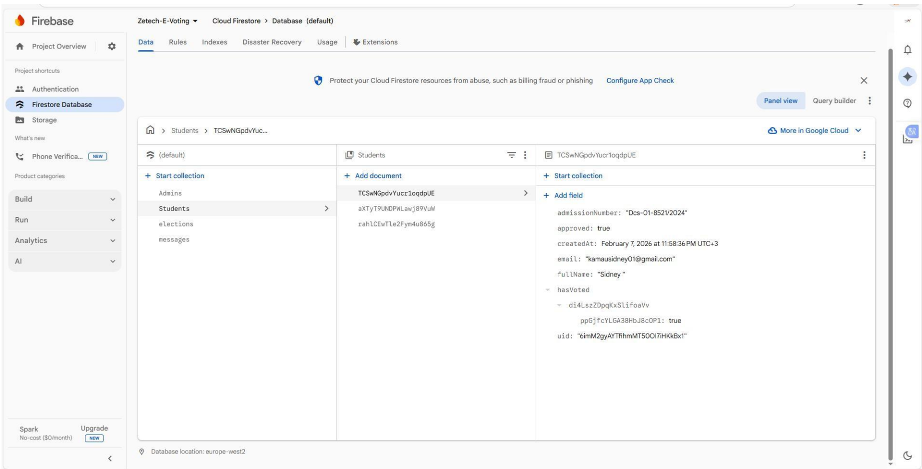


Figure: Screenshot of the logic implementation code.

3.3.3 Vote Casting and One-Vote Enforcement

This section explains how the system processes this functionality. The backend logic ensures security, accuracy, and reliability of the voting process.

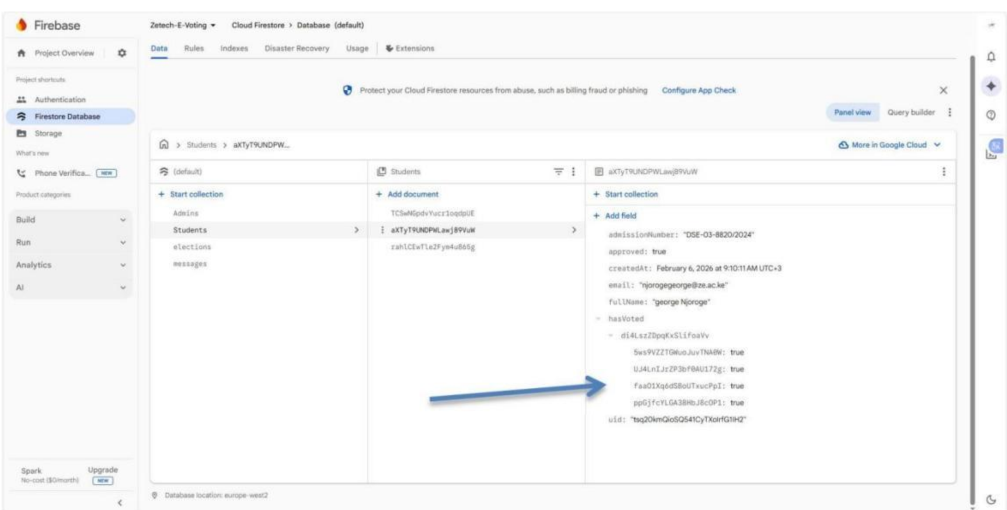


Figure: Screenshot of the logic implementation code.

3.3.4 Automated Vote Counting

This section explains how the system processes this functionality. The backend logic ensures security, accuracy, and reliability of the voting process.

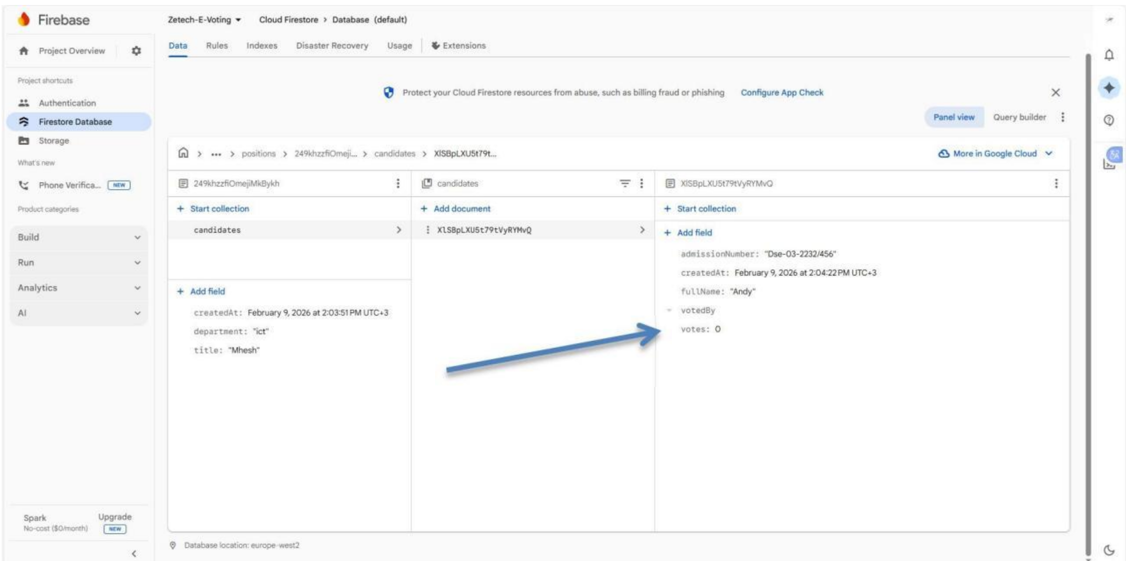


Figure: Screenshot of the logic implementation code.

3.3.5 Real-Time Results Display

This section explains how the system processes this functionality. The backend logic ensures security, accuracy, and reliability of the voting process.

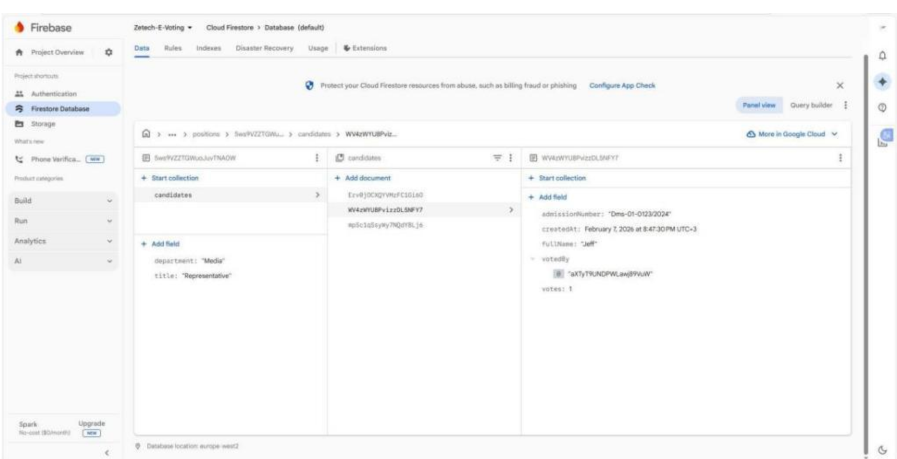


Figure: Screenshot of the logic implementation code.

3.4 Testing

The system was tested to ensure all features function correctly and securely. The following testing methods were conducted:

Unit Testing – Individual components such as login and vote casting were tested independently.

Integration Testing – Verified interaction between frontend and Firebase database.

User Acceptance Testing – Selected users tested the system and provided feedback.

Security Testing – Tested authentication, voter eligibility checks, and one-vote enforcement.

Performance Testing – Tested system response time under multiple login attempts.

During testing, minor errors such as incorrect data validation and UI responsiveness issues were identified and corrected to ensure smooth functionality.

3.5 Deployment

The web-based voting system was deployed using Firebase Hosting in conjunction with GitHub for version control and continuous deployment. The deployment process involved the following steps:

Firebase Project Configuration – A Firebase project was created and configured with enabled services including Authentication for user access control and Realtime Database for dynamic data storage.
Local Environment Setup – The local project directory was initialized and connected to the Firebase project using the Firebase Command Line Interface (CLI).
Build and Optimization – The application was built, and frontend files were optimized for production to ensure fast loading and efficient performance.
Deployment Execution – The system was deployed to Firebase Hosting using the command: `firebase deploy`.
Live URL Generation – Upon successful deployment, Firebase generated a live hosting URL, making the system accessible online via web browsers.

Once deployed, the system became fully operational and could be accessed by students and administrators on both desktop and mobile devices through any standard web browser.

CHAPTER FOUR: CONCLUSION AND RECOMMENDATIONS

4.1 Conclusion

The project successfully delivered a secure, accessible web-based voting system for Zetech University, addressing the inefficiencies of manual voting. The system was designed with Firebase (Authentication, Realtime Database) and deployed via Firebase Hosting with GitHub integration. Key lessons included hands-on experience with Firebase, frontend optimization, and the importance of user-centered design. Challenges such as real-time tallying latency and duplicate voting prevention were overcome through iterative testing and problem-solving.

4.2 Recommendations

To enhance the system, future developers should consider biometric verification to strengthen voter identity and blockchain audit trails for transparency and immutability. These upgrades would build on the current system to create a more robust and inclusive electoral platform.

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